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Improved methane production of corn straw by the stimulation of calcium peroxide

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ABSTRACT

This research focused on the effect of calcium peroxide (CaO_2) addition on anaerobic digestion (AD) of corn straw. It was found that the CaO_2 dose of 0.032–0.256 $g \cdot g^{-1}$ TVS (total VS of substrate and inoculum) had positive effect on the AD of corn straw by improving methane yield and content in biogas. Maximum methane yield was achieved at the CaO_2 dose of 0.032 $g \cdot g^{-1}$ TVS, which was 241.6 $mL \cdot g^{-1}$ VS_S (VS of substrate) and 11% higher than that of the control. Methane content in biogas also increased according to the in situ immobilization of CO_2 by $Ca(OH)_2$ that produced via the reaction of CaO_2 with water (XRD analysis results). Microbial community structure was significantly affected by CaO_2 addition. The variation of bacterial community structure led to higher hydrolysis rate, and the variation of archaea community structure ensured the proceeding of methanogenesis process. The use of CaO_2 could be a cost-effective method to improve methane production of corn straw because it avoids harsh condition and high cost (e.g., chemical and energy inputs) associated with conventional chemical treatment techniques. The digestate from the AD with the addition of CaO_2 can contain high $CaCO_3$ content, making it a potential soil amendment for acidic soil.

1. Introduction

Using cellulosic biomass as renewable resource to produce bioenergy has attracted global interests during the past decades [1,2]. Anaerobic digestion (AD) technology has been widely used for organic waste disposal and energy production [3]. In AD process, the cellulosic biomass can be converted into biogas, a mixture of methane, carbon dioxide and small amounts of other gases [4,5]. Biogas typically has a calorific value of 22,000–25,000 $kJ \cdot m^{-3}$ [6], therefore it is considered as one of the major renewable energy source. In general, AD process can be divided into four stages: hydrolysis, acidogenesis, acetogenesis, and methanogenesis [7]. Cellulosic biomass is mainly composed of cellulose, hemicellulose and lignin, and they bind with each other tightly by non-covalent forces and covalent cross-linkage [8]. Due to the complex components and strong structure of cellulosic biomass, the hydrolysis was demonstrated to be the rate-limiting step of AD [9,10], resulting in some common issues such as long retention time and low efficiency. Substrate pretreatment is the most studied method to improve the hydrolysis rate [7,10], including chemical, thermal, mechanical, and biological pretreatment as well as their combination. During pretreatment, the structure of substrate is partly destroyed. Which makes the substrate more accessible to microorganisms and extracellular hydrolytic enzymes present in AD. However, the pretreatment process is usually conducted in a separate reactor with the input of additional energy and/or chemicals, which leads to increased cost. Especially, the chemical pretreatment might also cause environmental problems. As a result, an economically feasible and environmentally friendly method should be developed to improve the hydrolysis rate and the overall AD efficiency of cellulosic substrate.

Recent research has shown the hydrolysis stage of AD could be accelerated under microaerobic condition [11,12]. In addition, microaerobic treatment (microaerobic pretreatment or introduction of limited oxygen into the AD system) has been successfully applied to improve the methane yield in several studies [13–16]. The microaerobic condition was considered to support the growth of aerobic and/or facultative microorganisms, which stimulated the secretion of extracellular hydrolytic enzymes (e.g., cellulose and protease) [14]. The production of more such enzymes could improve the hydrolysis of cellulosic substrate and finally increase the AD efficiency. Compared with other methods, the micro-aeration in AD by using oxygen only needs limited amount of oxygen or air, which is simpler, cheaper and more attractive [15].

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Calcium peroxide (CaO_2), a commonly used oxidant has been extensively applied in agriculture, aquiculture, environmental restoration, and pharmaceuticals [17]. It can react with water to produce calcium hydroxide [$Ca(OH)_2$] and oxygen [17]. Due to this property, CaO_2 could be used as an alternative additive to provide oxygen for AD process and create a microaerobic condition to enhance the hydrolysis of substrate. Previous studies have demonstrated CaO_2 was an efficient additive for contaminant (e.g., phenolic endocrine disrupting compounds and polychlorinated biphenyls-containing electrical insulating oil) removal with the action of reactive oxygen species (ROS) derived from CaO_2 [17,18]. In the AD of active sludge, CaO_2 was used as an efficient additive to improve the activities of both hydrolytic enzymes and acid-forming enzymes, and therefore enhance the hydrolysis and acidification of active sludge [19]. In addition, $Ca(OH)_2$ produced via the reaction of CaO_2 with water could immobilize CO_2 in situ ($Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$), and therefore improve the methane content in biogas. Also, the produced $Ca(OH)_2$ or $CaCO_3$ could contribute additional alkalinity to the AD system, and therefore benefit the production of soluble chemical oxygen demand (SCOD) and volatile fatty acids (VFs) [20]. However, the $^{\cdot}OH$ generation from CaO_2 was found to have significant inhibition on methanogens [21]. In the study of Li et al. [19], the production of methane was inhibited with the CaO_2 dose in the range of 0.05–0.3 $g{\cdot}g^{-1}$ VSS (volatile suspended solid). Therefore, the dose of CaO_2 needs to be well controlled when it is used as an additive to enhance the AD efficiency. A critical dose should be determined between the stimulation effect on hydrolysis (due to O_2 and H_2O_2 generation from CaO_2) and the inhibition effect (arising from $^{\cdot}OH$ generation from CaO_2) on methanogenesis. In addition, the critical dose may vary between cellulosic feedstocks.

As a result, this study explored the possibility of using CaO_2 as an additive to increase the AD efficiency of corn straw. The effect of CaO_2 addition on the methane yield, methane concentration and SCOD (soluble chemical oxygen demand) production were analyzed. The dynamics of microbial community structure during the AD process were also investigated. The implementation of this study would provide a cost-effective solution to improve the AD performance on cellulosic feedstocks.

2. Materials and methods

2.1. Substrate and inoculum

Corn straw was collected from the field of PingDu, Qingdao, Shandong province, China. Before AD, the corn straw was air-dried to moisture content of 8% and chopped to particle size of about 0.5 cm. The inoculum used in this study was obtained from a mesophilic AD (at 35 °C) plant which uses cow dung and corn straw as substrates with hydraulic retention time of 35 days. Before used, the inoculum was filtered through a sieve with the opening of ten mesh. The characteristics of substrate and inoculum were shown in Table 1.

Table 1
Characteristics of substrate and inoculum.

Characteristics	Substrate (corn straw)	Inoculum
Moisture content (%)	8.5 ± 0.3	97.9 ± 0.1
Total solid (TS, %)	91.5 ± 0.3	2.1 ± 0.1
Volatile solid (VS, % of TS)	79.4 ± 0.4	71.3 ± 0.3
C (% of TS)	36.7 ± 1.6	–
N (% of TS)	1.3 ± 0.4	–
H (% of TS)	4.9 ± 0.4	–
C/N (% of TS)	28.2 ± 1.2	–

Note: "–" means this parameter was not tested.

2.2. Batch AD test

Mesophilic (37 °C) batch AD of corn straw was conducted in quadruplicate for a total fermentation time of 43 days. To start AD, 5.33 g (TS) corn straw, 2.67 g (TS) inoculum and different amount of CaO_2 were mixed in 300-mL glass bottles. The doses of CaO_2 were 0 (control), 0.032, 0.064, 0.128, and 0.256 $g{\cdot}g^{-1}$ TVS (total VS of substrate and inoculum), respectively (coded as C0, C1, C2, C3, and C4). A blank group containing inoculum only (without substrate or CaO_2) was also carried out to test the biogas production from inoculum. The working volumes for all AD batches were adjusted to 200 mL by adding nutrient solution as described in Angelidaki et al. [22]. All the bottles were flushed with N_2 for 5 min to remove the oxygen. After sealed with rubber stoppers, all the bottles were incubated in a shaking water bath at 37 °C with the shaking speed of 120 rpm.

2.3. Analysis of the fermentation residue by X-ray diffraction (XRD)

After AD, the digestates were collected and dried at 105 °C for 8 h followed by milling. The digestate powders were then subjected to $CaCO_3$ analysis with a wide angle X-ray diffraction (Bruker D8 Advance, Germany) at 40 kV and 40 mA in the scanning angle of 5–80° (2θ) at a scanning speed of 1.0°/min.

2.4. Analysis of AD performance

Biogas production was monitored in volume with water displacement method, and the methane content in biogas was analyzed with a gas chromatograph (GC, SP 6890, Shandong Lunan Inc., China) using argon as carrier gas. The GC was equipped with a Porapak Q stainless steel analytical column (180 cm long and 3 mm OD) and a thermal conductivity detector. The analysis temperatures of the injector, detector and oven were set at 50, 100 and 80 °C, respectively. Spectrophotometric method was used for SCOD measurement [23]. Prior to the SCOD analysis, AD effluent samples were centrifuged for 5 min at 10,000 rpm and the supernatants were filtered through 0.45 μm glass microfiber paper [23]. TS and VS of the substrate and inoculum were measured by following the standard method [24]. The element contents of corn straw were quantified with an elemental analyzer (Vario EL III elemental analyzer).

2.5. Analysis of the microbial community structure

During AD process, aliquots of 8 mL effluent were collected from all the groups (C0-C4) at the 3rd, 12th and 40th day of AD. As C1 showed the highest methane production, it was selected as an example from the treatment groups (C1-C4) to compare with C0 for analyzing the effect of CaO_2 addition on the microbial community structure. The collected effluent samples were stored in a -80 °C freezer until used. In this study, genomic DNA was extracted using the E.Z.N.A.® DNA Kit (Omega Biotek, Norcross, GA, U.S.) according to manufacturer's instruction. One percent agarose gel electrophoresis and spectrophotometry (optical density at 260 nm/280 nm ratio) were used to check the quality of extracted DNA. The V3–V4 hypervariable regions of the 16S rRNA gene were subjected to high-throughput sequencing which was done by the Beijing Allwegene Tech, Ltd (Beijing, China) using the Illumina Miseq PE300 sequencing platform (Illumina, Inc., CA, USA). The V3–V4 regions of the bacterial 16S rRNA gene were amplified by using the universal forward primer 338F (5′-ACTCCTACGGGAGGCAGCAG-3) and reverse primer 806R (5′-GACTACHVGGGTWTCTAAT-3′).

These primers contained a set of 8-nucleotide barcodes sequence unique to each sample. The PCR program included steps of 95 °C for 5 min, 25 cycles at 95 °C for 30 s, 55 °C for 30 s, and 72 °C for 30 s as well as a final extension of 72 °C for 10 min. The PCR reactions were performed in triplicate in 25 μL mixtures containing 2.5 μL of 10X Pyrobest Buffer, 2 μL of 2.5 mM dNTPs, 1 μL of each primer (10 μM), 0.4

U of Pyrobest DNA Polymerase (TaKaRa), and 15 ng of template DNA. The amplicon mixture was applied to the MiSeq Genome Sequencer (Illumina, San Diego, CA, USA). Amplicons were extracted from 2% agarose gels, purified using the AxyPrep DNA Gel Extraction Kit (Axygen Biosciences, Union City, CA, U.S.) according to the manufacturer's instructions, and quantified using QuantiFluor™-ST (Promega, U.S.). Purified amplicons were pooled in equimolar and paired-end sequenced (2 × 300) on an Illumina MiSeq platform according to the standard protocols.

The extraction of high-quality sequences was performed with the QIIME package (Quantitative Insights Into Microbial Ecology) (v1.2.1). Raw sequences were then screened based on sequence length, quality, primer, and tag. Low-quality sequences were removed according to the following rules: (i) raw reads were shorter than 110 nucleotides; (ii) the 300 bp reads were truncated at any site receiving an average quality score < 20 over a 50 bp sliding window, and the truncated reads that were shorter than 50 bp were discarded; (iii) exact barcode matching: 2 nucleotide mismatch and reads containing ambiguous characters were removed; and (iv) only sequences whose overlap were longer than 10 bp were assembled according to their overlap sequence, and reads which could not be assembled were discarded.

The unique sequence set was classified into operational taxonomic units (OTUs) under the threshold of 97% identity with UCLUST. Chimeric sequences were identified and removed using Usearch (V8.0.1623). The taxonomy of each 16S rRNA gene sequence was analyzed by using UCLUST against the Silva119 16S rRNA database with the confidence threshold of 90%.

3. Results and discussion

3.1. Effect of CaO_2 on total gas production and methane content

As shown in Table 2, the addition of CaO_2 affected total biogas/methane yields and methane content differently. The CaO_2 dose of 0.032 $g{\cdot}g^{-1}$ TVS (C1) achieved the maximum biogas (476 $mL{\cdot}g^{-1}$ VS_s) and methane (241.6 $mL{\cdot}g^{-1}$ VS_s) yields. Further increase of CaO_2 dose caused significant decrease of total biogas yield compared to the control (C0), especially when CaO_2 doses were 0.128 (C3) and 0.256 (C4) $g{\cdot}g^{-1}$ TVS. All CaO_2 dose levels (C1–C4) achieved higher cumulative methane yield over C0 while further increase of CaO_2 dose from C1 to C4 slightly reduced the cumulative methane yield. With the addition of CaO_2, the methane content was significantly increased from 49.4 to 66.8%, and higher CaO_2 dose resulted in higher methane content, which could be one of the reasons why groups C3 and C4 still had higher cumulative methane yield than C0 even though their cumulative biogas yield was lower. The calorific value of biogas derived from methane component in biogas, thus higher methane content can lead to a higher calorific value of biogas. Biogas upgrading to increase the methane purity is an indispensable step prior to its utilization. Therefore, high methane content in biogas is beneficial for the reduction of energy consumption and cost for biogas upgrading and its final applications.

The microaerobic condition resulted from the presence of CaO_2 in AD could largely contribute to the increased cumulative methane yield. The added CaO_2 can react with water and release oxygen during AD

Table 2
Total biogas and methane production from AD of corn straw.

Groups	Cumulative biogas yield ($mL{\cdot}g^{-1}$ VS_S)	Cumulative methane yield ($mL{\cdot}g^{-1}$ VS_S)	Average methane concentration in biogas (%)
C0	438.8 ± 23.7	216.8 ± 8.5	49.4 ± 0.8
C1	476.4 ± 18.7	241.6 ± 8.1	50.7 ± 0.3
C2	441.7 ± 22.7	235.6 ± 10.5	53.4 ± 1.6
C3	422.8 ± 18.4	233.7 ± 7.7	55.3 ± 0.6
C4	346.2 ± 17.0	231.2 ± 6.5	66.8 ± 1.5

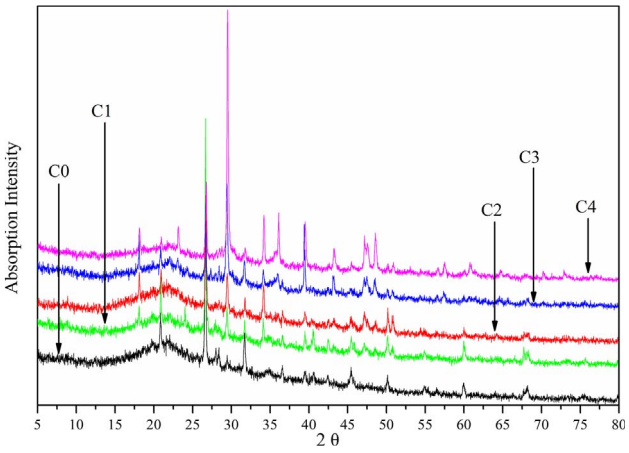


Fig. 1. XRD analysis results of fermentation residue.

($2CaO_2 + 2H_2O \rightarrow 2Ca(OH)_2 + O_2\uparrow$) [18]. The produced oxygen can induce the growth of facultative bacteria that produced more extracellular enzymes and improved the hydrolysis efficiency, leading to increased methane yield [11,25,26]. The CO_2 in the biogas could be partly immobilized in situ by $Ca(OH)_2$ according to the following reaction: [$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$]. Biogas is mainly composed by methane and carbon dioxide. The methane content in biogas is calculated according to the following formula: methane content in biogas = methane/biogas. The produced $Ca(OH)_2$ could partly immobilize carbon dioxide with no effect on methane. The amount of biogas decreased due to the immobilization of carbon dioxide, however the amount of methane did not change, which led to increased methane content in biogas. In addition, the produced $Ca(OH)_2$ and $CaCO_3$ could improve the acidic buffer capacity of the AD system, which could benefit the production of SCOD and volatile fatty acids (VFAs), and therefore resulting in higher methane yield and content [20].

In XRD analysis of AD digestates (Fig. 1), it was found that all AD digestates with the addition of CaO_2 (C1-C4) had much stronger $CaCO_3$ peak signals (at $2\theta = 29^{\circ}$) than that with no CaO_2 addition (C0). Moreover, the intensity of $CaCO_3$ peaks increased with the increase of CaO_2 dose. These results verified the fact that the added CaO_2 into the AD formed $CaCO_3$ through reactions of $2CaO_2 + 2H_2O \rightarrow 2Ca(OH)_2 + O_2\uparrow$ and $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$. Given the potential application of AD digestate as organic fertilizer for agriculture, the presence of $CaCO_3$ in the digestate could be beneficial for its use as a soil amendment for acidic soil

3.2. Effect of CaO_2 on daily and cumulative methane yields

As shown in Fig. 2A, the daily methane yields of C0-C2 followed the same trend with the peak value at the 5th day of AD. However, C1 and C2 had higher daily methane yield than C0 after the 2nd day of AD, which may be due to the micro-aeration and/or in situ CO_2 immobilization derived from the presence of CaO_2. In contrast, the daily methane yields of C3 and C4 were lower than that of C0 until the 6th and 10th day of AD, respectively, and their peaks delayed to the 8th and 12th day of AD, respectively. As reported in previous studies, the added CaO_2 could react with water and produced hydroxyl radical ($^{\cdot}OH$) [21], which was an inhibitor for methanogens [19]. In this study, the higher CaO_2 doses in C3 and C4 may generate more $^{\cdot}OH$ in AD and inhibited methane production to certain extent, leading to late occurrence of peak daily methane yield. Nevertheless, the daily methane yields of C3 and C4 exceeded that of C0 in the late stage of AD, which may indicate that the benefit of $CaOH_2$ addition overwhelmed its inhibition effect.

The timely cumulative methane yield also reflects the benefit of CaO_2 for AD (Fig. 2B), i.e., the final total methane yields for all CaO_2 treatment groups (C1-C4) are higher than that of the control group (C0)

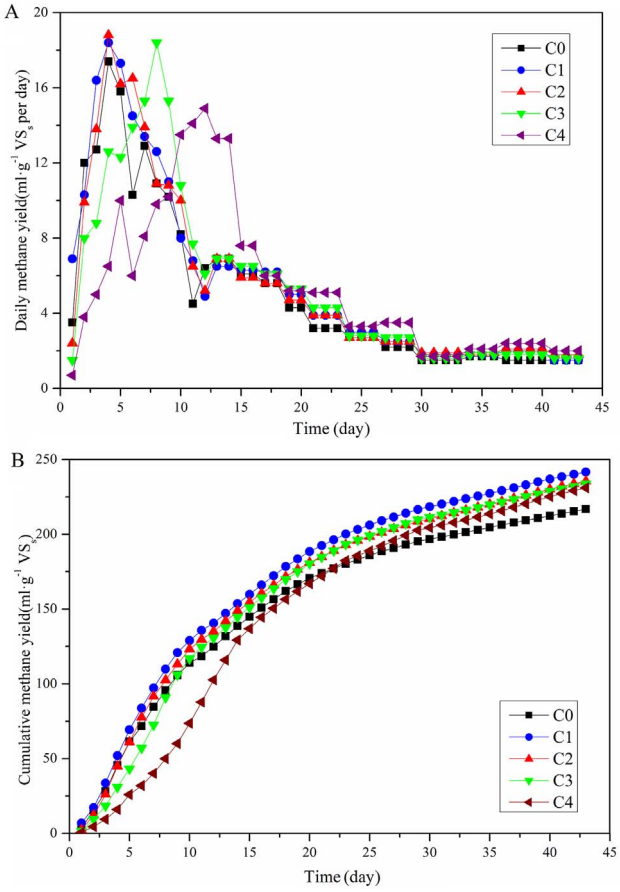


Fig. 2. Daily and cumulative methane yields from the AD of corn straw (A: daily methane yield and B: cumulative methane yield).

and rank in the order of C1 > C2 > C3 > C4 with the increase of CaO_2 dose. The maximum final methane yield (241.6 mL·g^{-1} VS$_s$) was obtained with the CaO_2 doses of 0.032 g·g^{-1} TVS (C1), which was 11% higher than that of C0. The groups of C1 and C2 had higher cumulative methane yield than C0 during the whole AD process, and the increments enlarged with the time. On the contrary, the cumulative methane yields of C3 and C4 were not always higher than that of C0 until the 11th and 23rd day of AD.

Improved AD performance by the addition of CaO_2 could be explained as follows: (1) added CaO_2 could reacted with water and released O_2 [19,27], which supported the growth of facultative/aerobic bacteria and induced the production of more extracellular enzymes to increase hydrolysis rate; (2) H_2O_2/˙OH were the intermediates of the reaction between CaO_2 and H_2O, and could act on the cellulosic substrate to destroy its structure, making it more accessible to biological degradation during AD; and (3) the produced $Ca(OH)_2$ and $CaCO_3$ could contribute to the alkalinity resulting in higher acid buffer capacity of the AD system, which could benefit for the hydrolysis of substrate and lead to more SCOD and volatile fatty acids (VFAs) production, and therefore provide more available substrate for methane production [20]. The overdose of CaO_2 can inhibit methane production (e.g., C3/C4 vs C0 in the early stage of AD), which might attribute to the inhibition of methanogens by excessive ˙OH and O_2 released from the reaction of CaO_2 with water. According to Fig. 2A/B and Table 1, the benefits of CaO_2 could include both micro-aeration and in situ CO_2 immobilization. The significance of these two benefits could switch depending on the dose of CaO_2. Therefore, using CaO_2 as an additive to improve the AD performance, the dosage is critical to maximize its benefits.

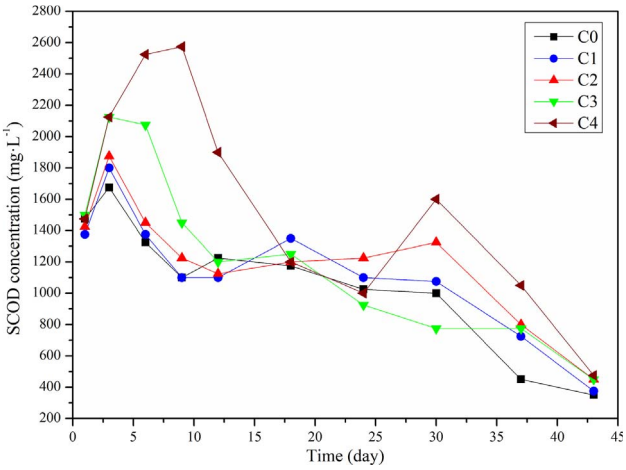


Fig. 3. Variation of SCOD concentration during AD of corn straw.

3.3. Effect of CaO_2 on the SCOD concentration during the AD of corn straw

The overall variations of SCOD concentration in AD process for all groups (C0-C4) followed the same trend, i.e., initial peaking, gradual decrease in mid-stage and final rapid decrease to a similar value (Fig. 3). The SCOD concentration showed a dynamic balance between SCOD production from hydrolysis/acidification process and subsequent consumption by methanogens. As shown in Fig. 3, the addition of CaO_2 had significant effect on the SCOD concentration during AD, i.e., the groups (C1–C4) with CaO_2 addition generally achieved higher SCOD concentrations than C0. The highest SCOD concentrations for different groups positively correlated with the CaO_2 doses. The maximum SCOD for C4 was 2575 mg·L^{-1} (53.7% higher than that of C0). Compared with C0, C1 and C2 had slightly higher SCOD concentrations and consumption (reflected by their similar final SCOD concentration) as a whole, which indicated the higher hydrolysis/acidification efficiency in C1 and C2, thus achieving higher daily and cumulative methane yields as shown in Fig. 2. For C3 and C4, the SCOD accumulation was found at the initial stage of AD (Fig. 3) due to the inhibition of methanogens, which reduced methane production (Fig. 2). Afterwards, the produced SCOD was consumed rapidly (Fig. 3), leading to a higher methane yield (shown in Fig. 2) probably because the inhibitor ˙OH was consumed for CO_2 immobilization, simultaneously mitigating the inhibition on methanogens and reducing CO_2 content in biogas. Treating waste active sludge, Li et al. [19] also found the use of CaO_2 increased the SCOD concentration significantly during anaerobic fermentation, i.e., 61% higher SCOD concentration was achieved with the CaO_2 dose of 0.3 g·g^{-1} VS$_s$, compared to the control with no CaO_2 addition.

3.4. Analysis of the microbial community structure of AD

The effect of CaO_2 on the microbial community structure of AD was reflected by the Shannon diversity index and relative abundance of sequence at phylum level (Figs. 4 and 5). During AD, the Shannon diversity indexes of both C0 and C1 increased with time. The Shannon diversity indexes of C0 at the 3rd, 12th and 40th day of AD were 3.93, 4.12 and 4.75, respectively, while they were 3.86, 4.60 and 4.70, respectively for C1 (Fig. 4). In comparison, the Shannon diversity indexes of C0 at the 3rd and 40th day were slightly higher than those of C1, while the Shannon diversity index of C0 at the 12th day was lower than that of C1. One of the reasons behind this result may be that the micro-aeration generated by CaO_2 in C1 promoted the growth of aerobic and/or facultative microorganisms in AD, which led to higher microbial diversity in C1 at certain stage (e.g., the 12th day in this study) of AD. Similarly, Lim et al. [28] also reported the microbial diversity under microaerobic condition was significantly higher than that under

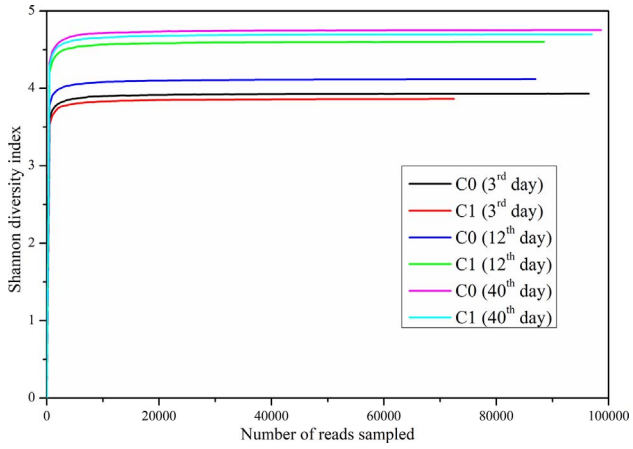


Fig. 4. Shannon diversity curves of C0 and C1 at the 3rd, 12th and 40th day of AD.

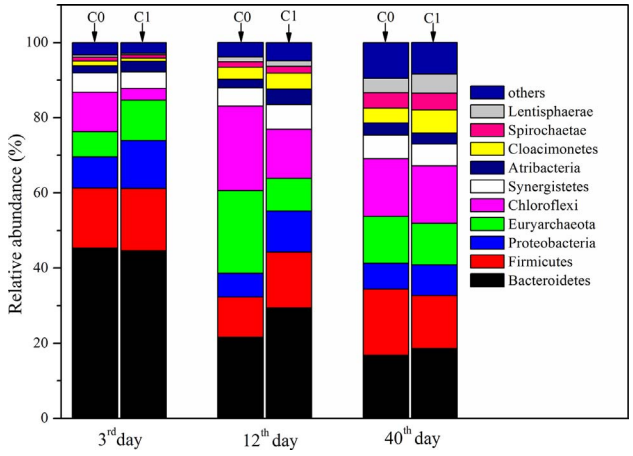


Fig. 5. Microbial community structures of C0 and C1 at phylum levels (phylum less than 1% of the total sequences and unclassified phylum were classified as others).

anaerobic condition.

As shown in Fig. 5, the microbial community structures of AD system changed versus time and were significantly affected by the addition of CaO_2. During the whole AD process, *Bacteroidetes* was the most dominant phylum followed by *Firmicutes*. *Proteobacteria*, *Euryarchaeota*, *Chloroflexi*, and *Synergistetes* were also among the dominant phyla. With the proceeding of AD, the relative abundances of *Bacteroidetes* and *Proteobacteria* decreased, while the relative abundances of *Atribacteria*, *Cloacimonetes*, *Spirochaetae*, and *Lentisphaerae* increased. The relative abundances of *Firmicutes* in C1 at the 3rd and 12th day of AD were 3.8% and 39.6% higher than those in C0, respectively. Microorganisms affiliating to the phylum *Firmicutes* were reported to be associated with the production of extracellular enzymes (e.g., cellulases and proteases), the degradation of organic compound and the formation of VFAs [29]. Therefore, the higher abundance of *Firmicutes* in C1 resulted in its stronger ability than C0 of degrading complex organic compounds. Similar to the results obtained in this study, Li et al. [16] also found that the relative abundance of *Firmicutes* increased significantly in the presence of CaO_2. Microorganisms belonging to phyla *Bacteroidetes* and *Proteobacteria* were reported to play crucial roles in hydrolysis and acetogenesis steps of AD [30,31]. Consequently, the higher relative abundances of *Bacteroidetes* and *Proteobacteria* in C1 than C0 yielded higher hydrolysis efficiency associated with higher SCOD in C1.

Conventionally, oxygen is considered lethal to methanogenic microorganisms [32]. However, some studies reported methanogens from genera *Methanobacterium* and *Methanosarcina* showed a certain oxygen tolerance capability [26,33]. It was found in this study that the relative abundances of genus *Methanobacterium* in C1 at the 3rd and 12th day of AD were much higher than those in C0 (Table S1), leading to higher methane yield in C1 than C0. Therefore, a certain level of micro-aeration in AD does not impair methanogenesis for methane production.

It appears that the variation of microbial community structure due to the CaO_2 addition is one of the main reasons for the improved AD performance on corn straw toward higher methane yield. The variation of bacterial community structure could lead to higher hydrolysis rate, and the variation of archaea community structure ensured the proceeding of methanogenesis process.

4. Conclusions

The addition of CaO_2 into the AD of corn straw improved the process performance significantly in increasing SCOD generation, methane yield and content in biogas as well as the gas production rate. Such benefits of CaO_2 could be resulted from its capabilities of: (i) improving hydrolysis efficiency of cellulosic substrate, (ii) increasing methane content in biogas by in situ CO_2 immobilization and (iii) providing additional buffer capacity for the AD system by contributing alkalinity. However, $^{\cdot}OH$ derived from the reaction of CaO_2 and H_2O inhibited methanogens to some extent. Therefore, the dose of CaO_2 needs to be well balanced between the beneficial and inhibition effect when it is used as an additive in AD process. In this study, the CaO_2 dose of $0.032\ g{\cdot}g^{-1}$ TVS achieved the optimum methane yield. The addition of CaO_2 seems to improve the microbial community structure of AD toward benefiting its performance. The utilization of CaO_2 as an additive could be a cost-effective technique to enhance the AD of cellulosic substrates.

Conflict of interest

The authors declare no competing financial interest.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.enconman.2018.02.070.

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